



Sizeamatic Pro

Stereo measurement software



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Stereo Length Measurement Method

A technical reference for interpreting Sizeamatic Pro stereo length measurements

Overview

Sizeamatic Pro calculates length measurements from calibrated stereo image pairs. The measured length is not calculated from the two dimensional pixel distance in either image. Instead, each measured endpoint is reconstructed as a three dimensional point from matched left and right image coordinates. The final length is then calculated as the straight line distance between two reconstructed three dimensional endpoints.

- Calibrate the stereo camera system.
- Rectify the left and right images using the calibration.
- Select the same physical feature in the left and right images.
- Reconstruct the selected feature as a three dimensional point.
- Repeat the reconstruction for a second endpoint.
- Calculate the length as the three dimensional distance between the two endpoints.

Calibration data

Before Sizeamatic Pro can make a stereo length measurement, the stereo camera system must be calibrated. The calibration process creates the files needed for measurement. These files include the left and right projection matrices used to relate three dimensional coordinates to rectified image coordinates.

The measurement process requires:

- Rectified left and right images
- A left projection matrix, written as P_L
- A right projection matrix, written as P_R
- Matched image points selected in the left and right images
- A known calibration unit, normally millimeters

The projection matrices are produced by the calibration process. They describe the stereo camera geometry and provide the scale needed to reconstruct measured points in three dimensional space. If the calibration files are missing, incorrect, or do not match the camera setup used to collect the imagery, the reconstructed points and length measurements will not be reliable.

Measured image points

For each measured point, the user selects the same physical feature in both rectified images. The selected image coordinates are written as:

$$p_L = (x_L, y_L) \quad p_R = (x_R, y_R)$$

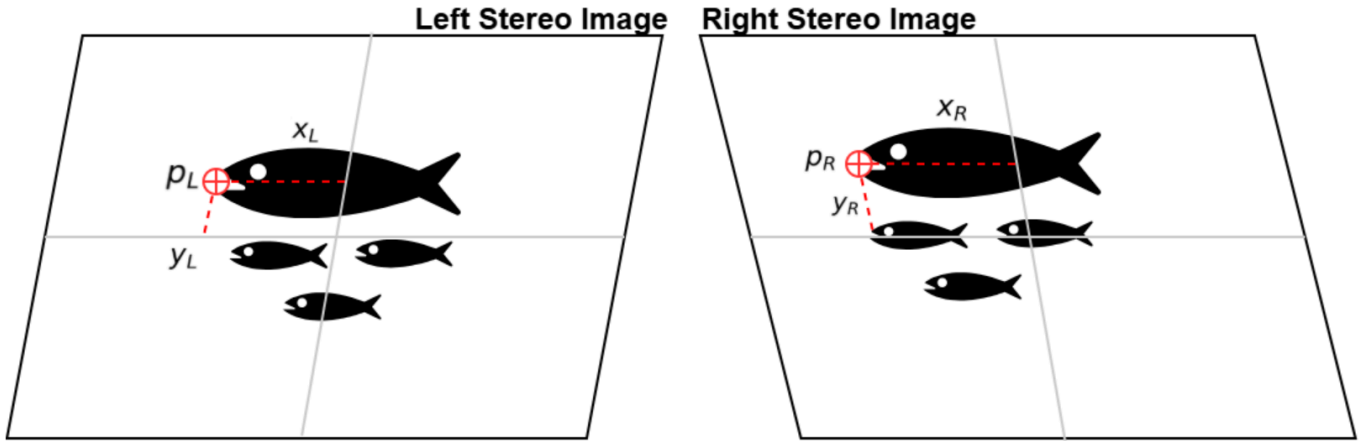


Figure 1. A Single Point is chosen from both Stereo Images

Because the stereo pair is rectified, matching points should lie on the same horizontal scanline. In practice, manual point selection can introduce a small vertical difference between the left and right selected points. Sizeamatic Pro uses the average of the two selected vertical coordinates:

$$y = \frac{y_L + y_R}{2}$$

The point pair used for reconstruction is then:

$$p_L = (x_L, y) \quad p_R = (x_R, y)$$

This reduces the effect of small vertical click differences in the left and right images.

Projection model

A calibrated camera projection matrix maps a three dimensional point into an image point. A three dimensional point is written in homogeneous form as:

$$X_h = [X \ Y \ Z \ 1]^T$$

For the left and right cameras, the projection equations are:

$$\lambda_L p_L = P_L X_h \quad \lambda_R p_R = P_R X_h$$

Here, P_L and P_R are the calibrated projection matrices. The terms λ_L and λ_R are image scale factors introduced by the homogeneous projection model. They are not measured directly. The points p_L and p_R are the measured image coordinates written in homogeneous form.

Three dimensional point reconstruction

For each matched image point, Sizeamatic Pro solves for the homogeneous three dimensional point that best satisfies the left and right projection equations:

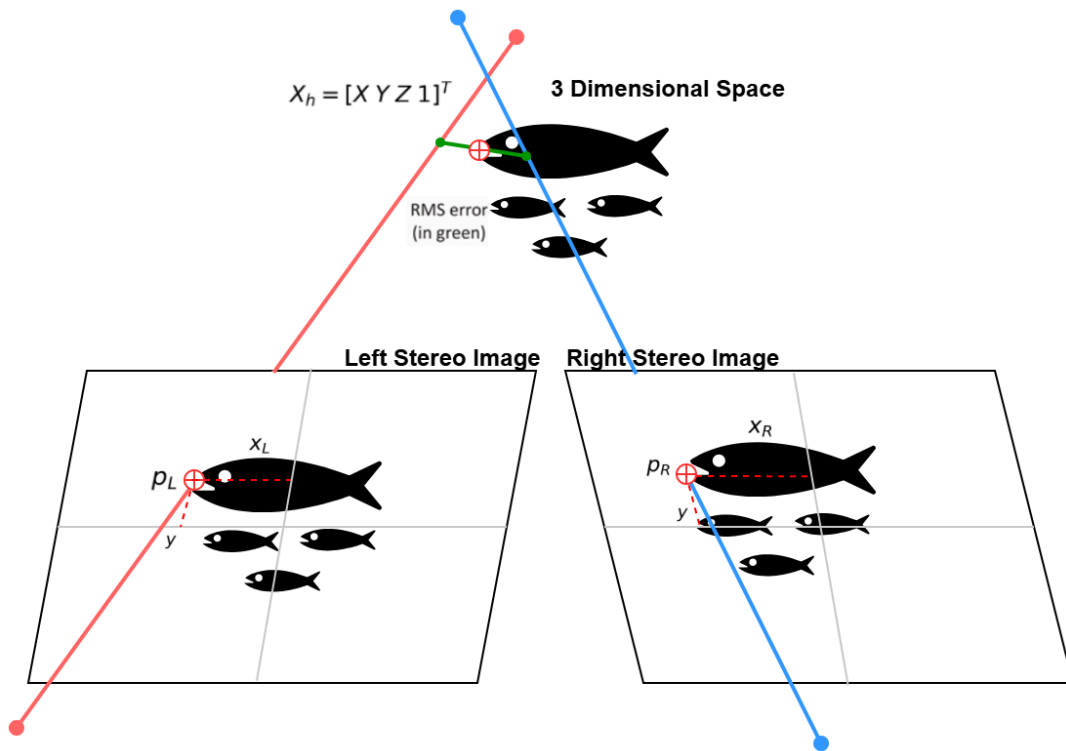


Figure 2. Screen coordinates are projected to 3D space using the Stereo Calibration Projection Matrices. RMS Error is calculated.

$$X_h = [X \ Y \ Z \ W]^T$$

One way to express the reconstruction problem is to write each projection matrix by its rows:

$$P = [p_1^T \ p_2^T \ p_3^T]^T$$

For a measured image point (x,y) , the projection constraint gives two independent linear equations:

$$(xp_3^T - p_1^T)X_h = 0 \quad (yp_3^T - p_2^T)X_h = 0$$

For a stereo point, the two equations from the left image and the two equations from the right image are stacked into one linear system:

$$AX_h = 0$$

The reconstructed point is the nonzero homogeneous vector X_h that best satisfies this system. Since real measurements contain small errors from calibration, image quality, synchronization, and point selection, the equations will usually not be satisfied perfectly. The solution is therefore the best fit homogeneous point for the measured stereo pair.

After the homogeneous point is reconstructed, Sizeamatic Pro converts it into normal three dimensional coordinates by dividing by W :

$$(X, Y, Z) = \left(\frac{X_h[0]}{W}, \frac{X_h[1]}{W}, \frac{X_h[2]}{W} \right)$$

The resulting coordinate is in the stereo calibration coordinate system, normally millimeters. If W is too close to zero, the point is rejected because the reconstructed position is unstable.

Length calculation

A length measurement uses two reconstructed three dimensional endpoints:

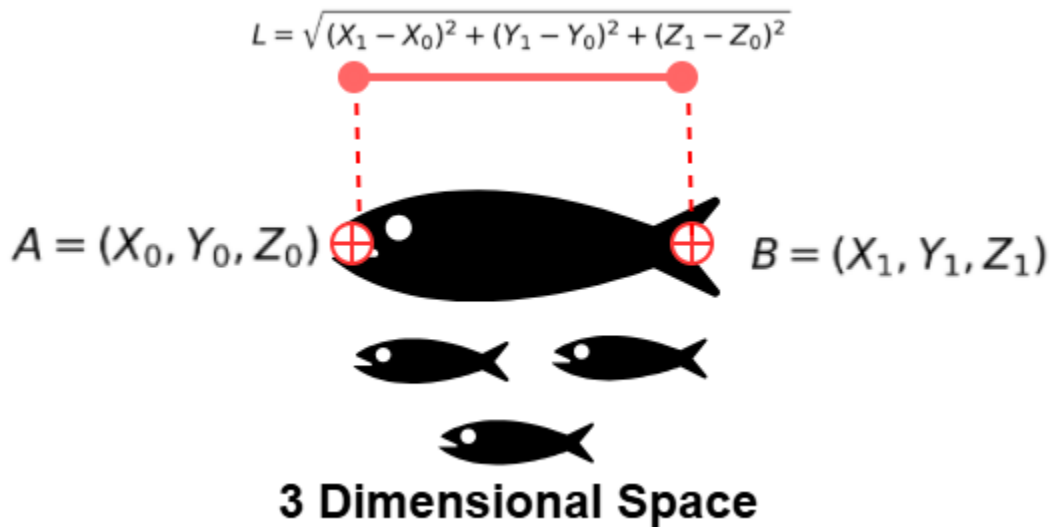


Figure 3. Calculated 3 Dimensional Length From Reconstructed 3D Points.

$$A = (X_0, Y_0, Z_0) \quad B = (X_1, Y_1, Z_1)$$

The coordinate differences are:

$$dX = X_1 - X_0 \quad dY = Y_1 - Y_0 \quad dZ = Z_1 - Z_0$$

The measured length is the three dimensional Euclidean distance:

$$L = \sqrt{(X_1 - X_0)^2 + (Y_1 - Y_0)^2 + (Z_1 - Z_0)^2}$$

or equivalently:

$$L = \sqrt{dX^2 + dY^2 + dZ^2}$$

This value is reported as Len(mm) in the segment output table.

Reprojection RMS

Sizeamatic Pro reports a reprojection RMS value for each reconstructed point. This is a pixel based consistency check. After a point has been reconstructed in three dimensional space, it is projected back into the left and right images using the same projection matrices.

For a reconstructed point X_h , the projected image coordinate is calculated as:

$$q = PX_h$$

The homogeneous image coordinate q is converted into normal pixel coordinates by dividing by its third component:

$$u = \frac{q_0}{q_2} \quad v = \frac{q_1}{q_2}$$

This gives the projected image point (u,v). The projected point is then compared with the original selected image point.

$$e_L = \sqrt{(u_L - x_L)^2 + (v_L - y_L)^2} \quad e_R = \sqrt{(u_R - x_R)^2 + (v_R - y_R)^2}$$

The reported reprojection RMS is:

$$RMS = \sqrt{\frac{e_L^2 + e_R^2}{2}}$$

This value is reported in pixels as ReprojRMS(px). A lower value means the reconstructed three dimensional point projects back closer to the selected image positions. A higher value can indicate poor point selection, poor image quality, calibration mismatch, camera movement after calibration, synchronization problems, or incorrect rectification.

Precision estimate

Sizeamatic Pro estimates local measurement precision by testing how sensitive the reconstruction is to small changes in selected image position. For each selected point, the image coordinates are shifted by a small pixel amount in the positive and negative directions. Each shifted point is reconstructed again. The spread of the resulting reconstructed values is used as a local estimate of uncertainty.

- SigmaZ(mm) is estimated from the spread of reconstructed Z values after small image coordinate perturbations.
- SigmaRange(mm) is estimated from the spread of reconstructed three dimensional range values after small image coordinate perturbations.
- SigmaLen(mm) is estimated from the spread of recalculated segment lengths after each endpoint is perturbed independently.

$$\Sigma_L = \text{std}(L_1, L_2, \dots, L_n)$$

This precision estimate describes local sensitivity to point selection. It is not a complete estimate of total measurement accuracy. It does not include every possible source of error, such as poor calibration, camera movement after calibration, incorrect calibration files, synchronization problems, or consistently selecting the wrong physical feature.

Measurement output fields

Sizeamatic Pro reports point values and segment values separately.

Points

Field	Meaning
idx	Point index.
X(mm)	Reconstructed three dimensional X coordinate.
Y(mm)	Reconstructed three dimensional Y coordinate.
Z(mm)	Reconstructed three dimensional Z coordinate, normally the main depth direction away from the stereo camera system.
Range(mm)	Three dimensional distance from the stereo coordinate origin to the reconstructed point.
Disp(px)	Stereo disparity in pixels, based on the horizontal difference between the selected left and right image coordinates.
dY(px)	Vertical difference between the selected left and right image coordinates. In a rectified stereo pair this should usually be small.
ReprojRMS(px)	Pixel reprojection RMS for the reconstructed point.
SigmaZ(mm)	Estimated local uncertainty in reconstructed Z caused by small image point changes.
SigmaRange(mm)	Estimated local uncertainty in reconstructed range caused by small image point changes.

Example point output

idx	X(mm)	Y(mm)	Z(mm)	Range(mm)	Disp(px)	dY(px)	ReprojRMS(px)	SigmaZ(mm)	SigmaRange(mm)
0	-378.0	49.9	696.4	794.0	-353.89	14.75	7.38	4.5	4.8
1	-188.0	13.9	593.1	622.3	-415.55	11.25	5.63	3.2	3.2

Segments

Field	Meaning
seg	Segment endpoints, such as 0 to 1.
dX(mm)	Difference in X between the two reconstructed endpoints.
dY(mm)	Difference in Y between the two reconstructed endpoints.
dZ(mm)	Difference in Z between the two reconstructed endpoints.
Len(mm)	Three dimensional Euclidean distance between the two reconstructed endpoints.
SigmaLen(mm)	Estimated local uncertainty in length caused by small image point changes.

Example segment output

seg	dX(mm)	dY(mm)	dZ(mm)	Len(mm)	SigmaLen(mm)
0 to 1	190.0	-36.0	-103.3	219.3	2.9

Interpreting measurement quality

Reprojection RMS and precision describe different aspects of a measurement. Reprojection RMS is a pixel consistency check. It describes how closely a reconstructed point projects back onto the original selected image points. A high reprojection RMS means the selected left and right image points, the calibration data, and the reconstructed point do not agree well.

Precision describes local repeatability or sensitivity. A low SigmaLen(mm) means that small changes in selected image position cause only small changes in the measured length. A high SigmaLen(mm) means the measured length is sensitive to small point placement changes.

A measurement can have low precision uncertainty but high reprojection RMS, or low reprojection RMS but still be affected by systematic error. For this reason, measurement quality should be evaluated using calibration quality, image quality, point selection quality, reprojection RMS, and the reported sigma values together.

High reprojection RMS can be caused by:

- Poor calibration
- Camera movement after calibration
- Incorrect calibration files
- Incorrect rectification
- Poor stereo synchronization
- Blur or poor image visibility
- Selecting different physical points
- Poorly defined biological landmarks

High length uncertainty can be caused by:

- Long range from the cameras
- Weak stereo geometry
- Measurements near the edge of the image
- Objects angled toward or away from the stereo pair
- Poorly defined endpoints
- Manual point placement uncertainty

Suggested methods wording for publications

Publication wording

Stereo length measurements were made using Sizeamatic Pro, developed by Marine Applied Research & Exploration. The stereo camera system was calibrated before measurement, producing the projection matrices required to relate rectified image coordinates to three dimensional coordinates. For each measured endpoint, the same physical feature was selected in the left and right rectified images. The matched image coordinates and calibrated projection matrices were used to reconstruct the corresponding three dimensional point in the stereo coordinate system.

Length was calculated as the Euclidean distance between two reconstructed endpoints, where the endpoint coordinates were expressed in millimeters. Point consistency was assessed using pixel reprojection RMS, calculated by projecting each reconstructed point back into the left and right images and comparing the reprojected positions with the original selected image points. Local measurement precision was estimated by perturbing selected image coordinates by a small pixel amount, reconstructing the points again, and calculating the resulting variation in depth, range, or length.

Validation note

This document describes how Sizeamatic Pro calculates stereo length measurements. Measurement confidence should also be supported by calibration reports and validation measurements of objects with known dimensions. Validation measurements should include objects measured at representative ranges, positions in the image, and orientations relative to the stereo camera system.